

ZOIKOSUITE

System Architecture Diagram Pack

Governed Business Operations Intelligence Platform

Classification

CONFIDENTIAL — INTERNAL STRATEGIC DOCUMENT

Standard

Enterprise SaaS · Architecture Blueprint

Architecture Style

Governance-First · Event-Driven · API-First · Multi-Entity · Multi-Jurisdiction

Version

1.1 — Sovereign Architecture Diagram Pack

Document

02 of 06 in the ZoikoSuite Architecture Series

ARCHITECTURE SERIES

- 01 Sovereign Back-End Architecture
- 02 System Architecture Diagram Pack (*this document*)
- 03 Microservices Specification Pack
- 04 Data Model / ERD Pack
- 05 Security Architecture Specification
- 06 Engineering Build Blueprint

01 · PURPOSE OF THIS DOCUMENT

This document translates the **ZoikoSuite Sovereign Back-End Architecture** into visual system topology.

It is the visual architecture counterpart to Document 01 and defines how the platform must be understood, communicated, and implemented across engineering, infrastructure, architecture, security, and enterprise stakeholder audiences.

The diagrams in this pack enable engineering teams, enterprise architects, infrastructure teams, institutional investors, and enterprise buyers to understand the platform at the correct level of depth:

- How the platform is structured — from client channels to the intelligence plane
- Where governance lives in the execution path — and why it cannot be bypassed
- How domain services are bounded and how they interact
- How data flows across the platform and which stores own which truth
- How events and evidence propagate through the system
- How identity, access, and entity context are enforced
- How infrastructure supports the platform at the Kubernetes, data, and cloud layers

This pack is not illustrative.

It defines the **actual system topology** engineering teams are expected to implement.

Each diagram is accompanied by **Diagram Notes** that clarify:

- architectural intent
- control assumptions
- non-negotiable constraints
- implementation consequences

02 · DIAGRAM INDEX

This pack contains eight architecture diagrams. Each communicates a distinct layer of the ZoikoSuite platform. Together, they form the complete visual blueprint for engineering, infrastructure, and enterprise architecture teams.

#	Diagram Title	Purpose
1	Platform Macro Architecture	Complete platform stack — client channels through intelligence plane
2	Governance Spine Architecture	Non-bypassable control layer between request and execution
3	Domain Services Architecture	Bounded domain topology across all operational domains
4	Event Architecture	Event-driven backbone for propagation, evidence, and intelligence

5	Data Architecture	Polyglot data model, routing, storage purpose, and source-truth rules
6	Identity & Access Architecture	Zero-trust authentication, authorization, and entity scoping
7	Evidence Architecture	Defensible evidence system — types, sources, properties, and stores
8	Infrastructure & Cloud Topology	Cloud-native deployment topology and supported isolation models

Implementation Note

This document defines actual platform topology. It is not conceptual artwork. Engineering teams are expected to implement in conformance with these architectural views.

DIAGRAM 1 · PLATFORM MACRO ARCHITECTURE

Complete system stack — from client channel to governed intelligence

CLIENTS & CHANNELS

Web Application · Mobile App · Admin Console · API Clients · Integration Partners



API GATEWAY / EDGE LAYER

Auth Enforcement · Rate Limiting · Schema Validation · Tenant Context Propagation · Distributed Tracing



IDENTITY & SESSION LAYER

SSO / SAML 2.0 · OAuth 2.0 / OIDC · MFA · Session Trust · Delegated Authority Resolution



◇ GOVERNANCE CONTROL PLANE ◇

Policy Engine

Jurisdiction Engine

Authorization Engine

Workflow & Approvals Engine

Obligations Engine

Evidence Requirements Engine

Decision Log

Non-bypassable — every request passes through this layer



DOMAIN EXECUTION SERVICES

Finance

Payroll

HR & Workforce

Legal & Contracts

Tax

Compliance

Commercial Ops



EVENT BACKBONE

Kafka / MSK — Durable · Ordered · Append-Only · Entity-Scoped · Jurisdiction-Aware



EVIDENCE SYSTEM

Audit Event Store

Document Vault

Policy Decision Log

Workflow History



DATA INFRASTRUCTURE

PostgreSQL / Aurora

Immutable Ledger Store

Redis / Valkey Cache

Object Storage

OpenSearch / Elasticsearch



ANALYTICS WAREHOUSE

Snowflake · BigQuery · Redshift

Board & Executive Reporting

Cross-Domain Analytics



INTELLIGENCE PLANE

Forecasting

Risk Detection

Compliance Gap Analysis

Anomaly Detection

Decision Support

Diagram Notes

- Every client and machine channel enters through the API Gateway. There is no uncontrolled ingress path.
- The Governance Control Plane sits between identity resolution and domain execution. This is the principal structural distinction of ZoikoSuite.

- The Event Backbone is the propagation and decoupling layer. Domain services emit facts; downstream consumers derive state without mutating source truth.
- Evidence is generated as a natural product of operations, not as a retrospective audit workflow.
- The Intelligence Plane is downstream of governed events and evidence. It never bypasses the Governance Plane.

DIAGRAM 2 · GOVERNANCE SPINE ARCHITECTURE

The non-bypassable control layer between every request and every material action

CLIENT REQUEST

Web · Mobile · API · Integration Partner · Scheduled Workflow



API GATEWAY / EDGE LAYER



IDENTITY RESOLUTION

Principal · Tenant · Legal Entity · Role Profile · Delegated Authority · Session Trust Context



◇ GOVERNANCE CONTROL PLANE ◇

NON-BYPASSABLE — every material action is evaluated here before execution

01 Policy Engine

Evaluates business, financial, legal, and internal control policies
Approval thresholds · Spend limits · Signatory matrices · SoD rules

02 Jurisdiction Engine

Determines regional or national rule sets
Payroll law · Tax rules · Termination constraints · Filing deadlines

03 Authorization Engine

Resolves whether the principal may act in the given tenant, entity, and workflow state

04 Workflow & Approval Engine

Converts governed actions into structured approval chains where auto-execution is not permitted

05 Obligations Engine

Maintains active statutory and policy obligations in real time

06 Evidence Requirements Engine

Determines what supporting evidence must exist before an action may proceed

07 Decision Logging Layer

Records every governance decision as immutable evidence

Subject · Actor · Rule set · Outcome · Timestamp · Approval reference



✓ AUTHORIZED

Action proceeds to domain service

X BLOCKED / ESCALATED

Denied, routed for approval, or escalated



DOMAIN SERVICE EXECUTES



DOMAIN EVENT EMITTED



EVIDENCE GENERATED

Diagram Notes

- The Governance Control Plane is the architectural spine of ZoikoSuite. It is not a module, not an optional feature, and not a policy overlay.
- No UI shortcut, internal API path, batch job, workflow runner, or external integration may circumvent this layer.
- Governance evaluation has only explicit outcomes: authorized, denied, or escalated. Silent bypass and silent failure are prohibited.
- Evidence begins at governance evaluation, not merely after transactional execution.
- This diagram is the single most important visual representation of the ZoikoSuite category thesis.

DIAGRAM 3 · DOMAIN SERVICES ARCHITECTURE

Bounded domains with explicit ownership under a unified governance spine

○ ALL DOMAINS OPERATE UNDER THE GOVERNANCE CONTROL PLANE ○



FINANCE DOMAIN

- General Ledger Service
- Accounts Payable / Receivable
- Treasury & Cash Position
- Bank Reconciliation Service
- Intercompany Accounting
- Consolidation Service
- Financial Close Service

PAYROLL DOMAIN

- Payroll Run Orchestrator
- Compensation Service
- Deductions Service
- Benefits Service
- Payroll Tax Service
- Employer Contributions Service
- Payroll Exceptions Service

HR & WORKFORCE DOMAIN

- Employee Master Service
- Onboarding Service
- Contract Issuance Service
- Leave & Absence Service
- Position & Organization Service
- Performance Review Service
- Offboarding Service

LEGAL & CONTRACTS DOMAIN

- Contract Lifecycle Service
- Clause & Template Service
- Obligation Tracking Service
- Board Resolution Service
- Corporate Actions Service
- Counterparty Management Service

TAX DOMAIN

- Tax Rules Service
- Tax Determination Service
- VAT / GST Engine
- Payroll Tax Engine
- Corporate Tax Estimation Service
- Withholding Tax Service
- Filing Preparation Service

COMPLIANCE DOMAIN

- Obligations Registry
- Deadline Engine

Filing Tracker
Compliance Status Service
Evidence Sufficiency Service
Exception & Escalation Service

COMMERCIAL OPS DOMAIN
Procurement Workflow Service
Purchase Request Service
Purchase Order Service
Invoice Approval Service
Vendor Due Diligence Service
Spend Controls Service

SHARED PLATFORM CAPABILITIES
Event Backbone · Evidence System · Identity Layer · Data Plane · Observability

Diagram Notes

- Each domain is bounded and must have explicit ownership, service contracts, and data responsibility.
- No domain self-authorizes material actions. Governance is externalized into the Governance Plane and enforced consistently.
- Domain interaction should prefer event-driven propagation over tight synchronous coupling.
- Shared platform capabilities are not owned by any single business domain. They are platform engineering responsibilities.
- This structure allows modular capability without compromising unified control.

DIAGRAM 4 · EVENT ARCHITECTURE

Event-driven backbone for propagation, evidence capture, and governed intelligence

EVENT PRODUCERS — DOMAIN SERVICES

Every domain service emits typed, governed events after authorized execution

Examples:

journal.posted
payment.initiated
period.closed
reconciliation.completed
payroll.run.completed

payroll.exception.raised
employee.hired
employee.terminated
contract.executed
obligation.created
tax.liability.updated
filing.submitted
obligation.overdue
compliance.gap.detected
exception.escalated
resolution.approved



EVENT BACKBONE

Kafka / MSK — Durable · Ordered · Partitioned by Entity · High-Throughput

Immutable

Events are facts, not commands

Append-Only

The event log is the historical record

Entity-Scoped

Every event carries tenant and legal-entity context

Jurisdiction-Aware

Jurisdiction context propagates with event



EVENT CONSUMERS

Evidence System

Captures every event as immutable evidence

Intelligence Plane

Consumes stream for anomaly detection, forecasting, and risk scoring

Analytics Warehouse

Feeds executive reporting and cross-domain analytics

Workflow Engine

Triggers approvals, escalations, and obligation follow-up

Observability System

Tracks platform health, SLA metrics, and business-event rates

Diagram Notes

- Event consumers must never mutate source domain truth.
- Every event payload must carry tenant ID, entity ID, actor identity, correlation ID, and jurisdiction context.
- The Event Backbone enables domain decoupling and governed propagation.
- The event log is itself part of the evidence model.
- Domain events are facts after execution, not instructions for upstream mutation.

DIAGRAM 5 · DATA ARCHITECTURE

Polyglot, fit-for-purpose data stores with strict routing and source-truth discipline

Architectural Principle

No single database solves every platform need. Each store exists for a specific purpose, under a specific routing rule, with a specific governance obligation.

OPERATIONAL RELATIONAL STORE

Technology

PostgreSQL / Aurora PostgreSQL-Compatible

Purpose

Primary source of truth for transactional domain data

Governing Rule

Authoritative operational store. Never bypassed by analytics or reporting layers.

Primary Uses

- Finance transactions and journal metadata
- HR and payroll records
- Contract and obligation metadata
- Compliance state and exceptions

IMMUTABLE LEDGER STORE

Technology

Append-Only Relational / Event-Sourced / Hash-Chained Ledger

Purpose

Tamper-resistant store for audit-critical financial and governed records

Governing Rule

Committed records may not be overwritten. Corrections occur only through explicit adjustments.

Primary Uses

- Journal entry ledger
- Financial close snapshots
- Payroll run archives
- Cryptographic integrity chains

DOCUMENT & EVIDENCE STORE**Technology**

Object Storage + Relational Metadata

Purpose

Version-controlled storage for contracts, resolutions, filings, payslips, and evidential artifacts

Governing Rule

Residency, retention, integrity, and access lineage must be policy-enforced.

Primary Uses

- Contract documents with version lineage
- Board resolution archives
- Filing artifacts
- Tamper-detection hashes

SEARCH LAYER**Technology**

OpenSearch / Elasticsearch

Purpose

Fast search and retrieval across documents, evidence, obligations, and legal artifacts

Governing Rule

Read-only derivative. Never a source of operational truth.

Primary Uses

- Audit retrieval

- Contract clause search
- Obligation lookup
- Evidence discovery

CACHE LAYER

Technology

Redis / Valkey

Purpose

Session acceleration, hot rule caching, and non-authoritative performance optimization

Governing Rule

Never authoritative. All business decisions validate against source stores.

Primary Uses

- Session and token caching
- Governance-rule hot cache
- API response acceleration
- Rate limiting and quota support

ANALYTICS WAREHOUSE

Technology

Snowflake / BigQuery / Redshift

Purpose

Read-optimized analytical store for executive reporting, forecasting, and cross-domain intelligence

Governing Rule

Strictly read-derived. Analytics may not mutate source operational truth.

Primary Uses

- Board and executive reporting
- Multi-entity reporting
- Anomaly and trend analysis
- Compliance gap analytics

Diagram Notes

- Operational truth remains in source domain systems.

- Entity and jurisdiction tags must persist throughout the data model and in all derived stores.
- Sensitive data such as payroll, HR, and PII must be field-classified and access-restricted.
- Historical records must preserve effective-dated rule context, not merely current-state interpretation.
- Data-store specialization is architectural, not incidental.

DIAGRAM 6 · IDENTITY & ACCESS ARCHITECTURE

Zero-trust, entity-scoped, deeply modeled access tied to governance at every layer

IDENTITY PROVIDERS

Enterprise IdP
SSO · SAML 2.0 · LDAP

API & Service Identity
OAuth 2.0 · OIDC · mTLS

Trust Controls
MFA · Device Trust · Session Limits



IDENTITY SERVICE — TOKEN VALIDATION & CONTEXT ASSEMBLY

Authenticated Principal · Tenant · Active Legal Entity · Role Profile · Delegated Authority · Session Trust



AUTHORIZATION ENGINE

RBAC
Defines broad access roles and functional reach

ABAC
Makes context-sensitive decisions using entity, jurisdiction, workflow state, action type, and risk level

SoD
Enforces segregation-of-duties constraints

Entity Scope

Hard-enforces legal-entity authority boundaries

Action Approval

Triggers explicit approval chains for high-risk actions



GOVERNANCE PLANE RECEIVES RESOLVED IDENTITY + AUTHORIZATION CONTEXT



ALL ACCESS DECISIONS ARE LOGGED

Successful · Denied · Escalated · Delegated

Diagram Notes

- Access is not a bolt-on. Identity and authorization are deeply modeled platform primitives.
- Zero-trust means every request is authenticated and re-evaluated, regardless of origin.
- SoD enforcement is architectural and runtime-enforced, not policy-only.
- Access decisions are evidential records and must be logged as such.
- Entity scope is mandatory, not optional, for every material action.

DIAGRAM 7 · EVIDENCE ARCHITECTURE

Defensible evidence as an architectural output, not an audit overlay

Most enterprise systems log activity. ZoikoSuite produces defensible evidence. That is a materially higher standard.

EVIDENCE SOURCES

User Actions

Service Events

Workflow Transitions

Document Versioning

External Integrations

Policy Evaluations

AI Recommendations & Overrides



EVIDENCE SYSTEM

Audit Event Store

Document Vault
Policy Decision Log
Workflow History Store
Approval Chain Archive
Evidence Manifest



EVIDENCE TYPES
Transaction Evidence
Approval Evidence
Document Evidence
Policy Decision Evidence
Workflow Evidence
Compliance Evidence
Audit Log Evidence

Required Properties of Every Evidential Record

- **Timestamped** — precision timestamps with full provenance chain
- **Actor-Bound** — the human or system actor is always identified
- **Entity-Bound** — every record is scoped to legal entity context
- **Jurisdiction-Aware** — rule basis and jurisdictional logic are retained
- **Immutable** — append-only and tamper-resistant by architecture
- **Source-Linked** — correlated to the originating action and event chain
- **Retrievable** — structured for audit, regulator, and legal discovery use

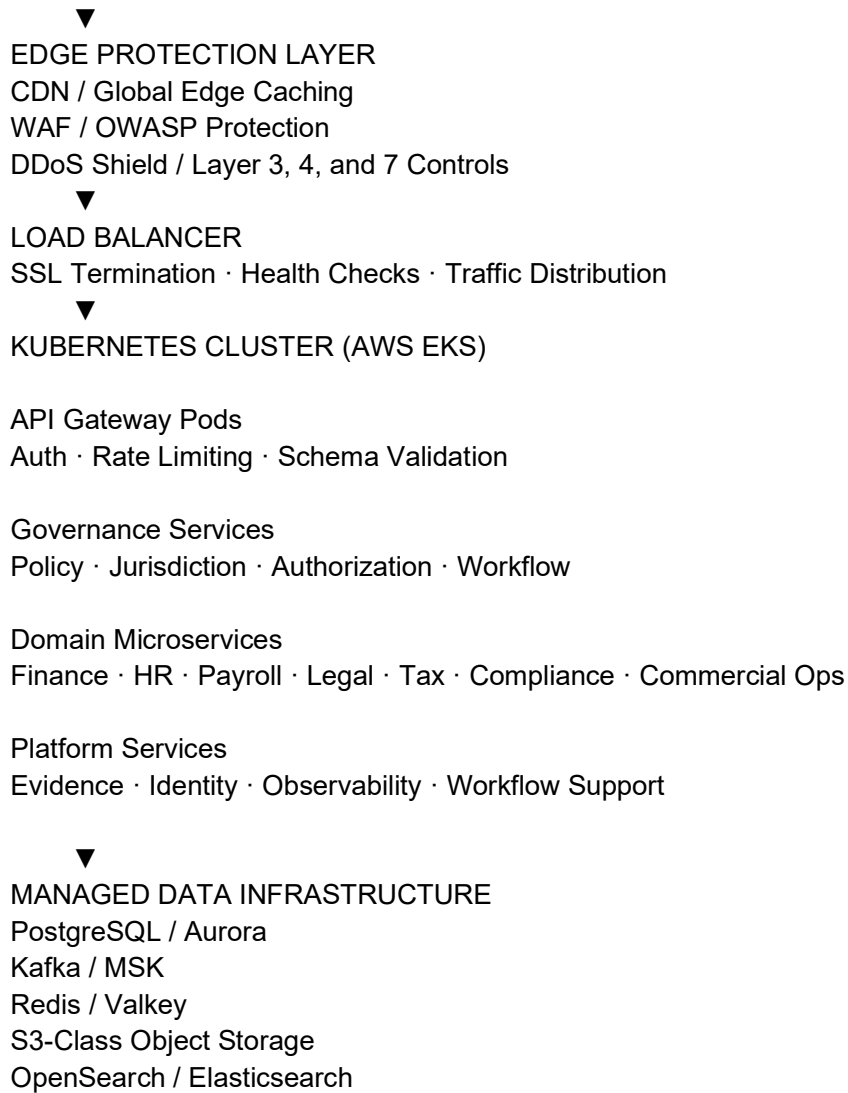
Diagram Notes

- Evidence is generated automatically as part of the governance and execution cycle.
- Every governance decision is itself evidence.
- The Document Vault must preserve version history, approval lineage, access history, and integrity validation.
- Evidence design must support regulator examination, internal compliance review, and legal discovery scenarios.
- Evidence is not merely stored; it is structured for defensibility.

DIAGRAM 8 · INFRASTRUCTURE & CLOUD TOPOLOGY

Cloud-native, Kubernetes-orchestrated, security-first deployment architecture

INTERNET / CLIENT INGRESS



Supported Deployment Modes

Deployment Mode	Description	Isolation Level
Multi-Tenant SaaS	Shared infrastructure with logical tenant isolation	Standard
Dedicated Private Cloud	Single-tenant logical isolation on shared substrate	Enhanced
Enterprise Single-Tenant	Dedicated infrastructure for compliance-mandated segregation	Highest

Sovereign / On-Premise

Customer-controlled
deployment for legal or
residency constraints

Sovereign Requirement

Diagram Notes

- AWS is the practical default, but the architecture should remain cloud-portable where commercially sensible.
- All workloads run under Kubernetes orchestration with network policy enforcement, service isolation, and controlled rollout discipline.
- Infrastructure as Code is mandatory. All platform changes must be version-controlled and audit-logged.
- Blue-green and canary patterns are preferred for zero-downtime deployment.
- Schema migrations must be backward-compatible by default.
- Deployment activity is part of the evidence record.

FINAL ARCHITECTURAL DOCTRINE

Every diagram in this pack exists to support one non-negotiable principle:

Business operations must execute inside governance.

Not around it.

Not after it.

Not as an audit overlay.

Inside it.

That is the ZoikoSuite architecture.

Each diagram in this pack is a different cross-section of the same system — the same doctrine expressed through topology, control flow, domain structure, event propagation, data routing, identity enforcement, evidence production, and infrastructure deployment.

NEXT DOCUMENT IN SERIES

Document 03 — Microservices Specification Pack

This document will define:

- every microservice
- service responsibilities and boundaries
- service APIs and contracts
- event outputs and schemas
- data ownership rules

- scaling and deployment models

It is one of the most important engineering documents in the ZoikoSuite platform series.